

Basic Laws of Boolean Algebra

Note that in Boolean Algebra, + represents OR and * represents AND

$A + (B + C) = (A + B) + C$ $A * (B * C) = (A * B) * C$	Associative laws	$A \ \ (B \ \ C) = (A \ \ B) \ \ C$ $A \ \&\& \ (B \ \&\& \ C) = (A \ \&\& \ B) \ \&\& \ C$
$A + B = B + A$ $A * B = B * A$	Commutative laws	$A \ \ B = B \ \ A$ $A \ \&\& \ B = B \ \&\& \ A$
$A(B + C) = AB + AC$ $A + BC = (A + B)(A + C)$	Distributive law for AND over OR Distributive law for OR over AND	$A \ \&\& \ (B \ \ C) = (A \ \&\& \ B) \ \ (A \ \&\& \ C)$ $A \ \ (B \ \&\& \ C) = (A \ \ B) \ \&\& \ (A \ \ C)$
$(A + B) * (C + D) = AB + AD + BC + BD$	FOILing AND over OR	$(A \ \ B) \ \&\& \ (C \ \ D) = (A \ \&\& \ C) \ \ (A \ \&\& \ D) \ \ (B \ \&\& \ C) \ \ (B \ \&\& \ D)$
$\overline{\overline{A}} = A$	Double negative law	$!(\neg A) = A$
$A + \overline{A} = 1$ $A * \overline{A} = 0$	Laws of Complement	$A \ \ \neg A = \text{true}$ $A \ \&\& \ \neg A = \text{false}$
$A + A = A$ $A * A = A$	Idempotence Laws	$A \ \ A = A$ $A \ \&\& \ A = A$
$A + 0 = A$ $A * 1 = A$	Identity Laws	$A \ \ \text{false} = A$ $A \ \&\& \ \text{true} = A$
$A + 1 = 1$ $A * 0 = 0$	Law of Union Law of Intersection	$A \ \ \text{true} = \text{true}$ $A \ \&\& \ \text{false} = \text{false}$
$\overline{A + B} = \overline{A} * \overline{B}$ $\overline{A * B} = \overline{A} + \overline{B}$	DeMorgan's laws	$\neg(A \ \ B) = \neg A \ \&\& \ \neg B$ $\neg(A \ \&\& \ B) = \neg A \ \ \neg B$
$A * (A + B) = A$ $A + A * B = A$	Laws of Absorption	$A \ \&\& \ (A \ \ B) = A$ $A \ \ (A \ \&\& \ B) = A$
$A + \overline{A} * B = A + B$	Law of Disappearing Opposite	$A \ \ \neg A \ \&\& \ B = A \ \ B$
$A \oplus B = A * \overline{B} + \overline{A} * B$ $A \oplus B = (A + B) * \overline{A * B}$	Exclusive OR (XOR function) a or b is true, but not both	$a \wedge b = (a \ \&\& \ !b) \ \ (b \ \&\& \ !a)$ $= (a \ \ b) \ \&\& \ \neg(a \ \&\& \ b)$
$\overline{A \oplus B} = (\overline{A} + B) * (A + \overline{B})$ $\overline{A \oplus B} = A * B + \overline{A} * \overline{B}$	Exclusive NOR function (XNOR function)	$\neg(a \wedge b) = (\neg a \ \ b) \ \&\& \ (a \ \ !b)$ $= (a \ \&\& \ b) \ \ (\neg a \ \&\& \ !b)$

Remember that: $\neg(a < b)$ is the same as $a \geq b$

	c = a & b			c = a b			c = a ^ b			c = !(a & b)			c = !(a b)			c = !(a ^ b)		
Truth Tables	a	b	c	a	b	c	a	b	c	a	b	c	a	b	c	a	b	c
	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	1
	0	1	0	0	1	1	0	1	1	0	1	1	0	1	0	0	1	0
	1	0	0	1	0	1	1	0	1	1	0	1	1	0	0	1	0	0
	1	1	1	1	1	1	1	1	0	1	1	0	1	1	0	1	1	1